

AIMO Proof Pilot — Final Grade

Problem (#_ID): 5_POLYGO

Submission: team_3

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (docs/markschemes.pdf).

1. Final mark

Final mark (choose one):

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. **ME1**) where possible.

Correct solution with converse checked correctly. A different approach to the mark scheme using the Rational Root Theorem and factorising $P(x) = (x - N)Q(x)$. A small slip-up when justifying the final choice of signs, which comes from evaluating $Q(1) = 0$ instead of the relations $(4f)$, but insufficient to constitute a deduction.